

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: T. Y. B. Tech.

Branch : Mechanical Engineering /

Semester : VI

Mechanical Engineering (Sandwich)

Subject Code & Name: BTMEC601_Y19 Manufacturing Processes- II

Max Marks: 60

Date: 30/06/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	In which of the grinding methods rate of material removal is high	Understand	1
	a. Surface Grinding b. Cylindrical Grinding c. Creep Feed Grinding d. All of the above		
2	Removing dull grains in order to make grinding wheel sharp is known as	Remember	1
	a. Loading b. Glazing c. Dressing d. Trueing		
3	An open structure of a grinding wheel is used for	Remember	1
	a. Soft materials b. Tough materials c. Ductile materials d. All of these		
4	The surface of the single point cutting tool on which the chips formed in cutting operation slide is called as	Remember	1
	a. Flank b. Heel c. Face d. Shank		
5	The shear angle is determine by the equation	Remember	1
	a. $\tan \phi = \frac{\cos \alpha}{1 - r \sin \alpha}$ b. $\tan \phi = \frac{r \cos \alpha}{1 - r \sin \alpha}$ c. $\tan \phi = \frac{r \cos \alpha}{1 - \sin \phi \alpha}$ d. $\tan \phi = \frac{\cos \alpha}{r - r \sin \alpha}$		
6	In Taylor's tool life equation $VT^n = C$ the exponent largely depends on	Understand	1
	a. Tool Material b. Cutting Condition c. Work Material d. None of these		
7	If t_1 denotes the uncut chip thickness and t_2 denotes cut chip thickness ratio then, chip thickness ratio ' r ' =	Understand	1
	a. $r = t_1/t_2$ b. $r = t_2/t_1$ c. $r = t_1 * t_2$ d. None of these		

8	In atomization				Understand	1
	a. passing molten metal through orifice and cooling it by dropping into water	b. molten metal is forced through a small orifice and broken up by a stream of compressed air	c. powder of metal is made by ball milling	d. the hydrogen reduces the oxide to metallic powder		
9	Cutting fluids are used to				Remember	1
	a. Cool the tool	b. Improve surface finish	c. Cool the workpiece	d. All of these		
10	A part produced by Powder metallurgy is termed a				Remember	1
	a. Welded part	b. Cast part	c. Forging part	d. Sintered part		
11	In compression moulding the curing time is				Understand	1
	a. Less	b. Moderate	c. More	d. None		
12	Blow moulding is a very slow process, however economical for producing products with better quality.				Understand	1
	a. True	b. False				
Q. 2	Solve the following.					12
A)	Explain in detail any two finishing operation in grinding.				Remember	6
B)	Explain three mechanisms of Grinding wheel wear.				Remember	6
Q.3	Solve the following.					12
A)	Derive mathematical expression to determine the shear angle in orthogonal cutting.					6
B)	Explain with fig. nomenclature of single point cutting tool.				Remember	6
Q. 4	Solve Any Two of the following.					12
A)	Explain properties and types of cutting fluid.				Remember	6
B)	What is tool life? Explain factors affecting on tool life.				Understand	6
C)	During the machining of certain compound using a H.S.S. tool at feed of 03 mm/min and depth of cut 2.5 mm the following observation were recorded when $V_1=30\text{m/min}$, $T_1=80\text{min}$, $V_2=35\text{m/min}$, $T_2=15\text{ min}$. Calculate n and c , the cutting speed for desired tool life of 50 min.				Applying	6

Q.5	Solve Any Two of the following.		12
A)	Define Powder Metallurgy. State limitation and advantage of Powder Metallurgy process.	Understand	6
B)	Describe the principle of centrifugal atomization with spinning disk with the help of schematic diagram	Understand	6
C)	Discuss the sintering mechanisms with the help of schematic diagram.	Understand	6
Q. 6	Solve Any Two of the following.		12
A)	Explain with neat sketch extrusion molding, advantages and limitation.	Remember	6
B)	Describe the blow moulding process with neat sketch.	Understand	6
C)	Describe with fig. Reaction Injection molding.	Understand	6
	*** End ***		